The Width of Aerosol Size Distribution: A Nonnegligible Factor in the Hyperspectral Infrared Radiative Transfer Simulations and Aerosol Remote Sensing

Jiaqi Zhao, Shuai Hu, Lei Liu, Ruijun Dang, Yao Xiao, and Yulong He

***Abstract*—Remote sensing of aerosols by infrared hyperspectral radiation has become an emerging research frontier in recent years. In current infrared hyperspectral remote sensing of aerosol, only the aerosol type and effective radius (R_{eff}) are considered, while the width of the aerosol size distribution geometric standard deviation (GSD) is simplified as a constant, neglecting its impact on infrared radiative transfer. Is this simplification physically justified? To clarify this issue, an infrared hyperspectral radiative transfer model considering the variation of GSD of aerosol particles was established, and the impacts of GSD on both upwelling and downwelling radiances are investigated. It can be found that radiance discrepancies attributed to variations of GSD exceed those caused by effective radius up to 70%. The downwelling radiance demonstrates far greater sensitivity to GSD compared to its upwelling counterpart. Furthermore, the sensitivity of infrared radiance to GSD will be amplified with the increase of aerosol optical depth (AOD). Finally, to further validate the necessity of considering GSD in aerosol remote sensing, both triple-parameter (AOD, R_{eff}, and GSD) and dual-parameter inversions (AOD and R_{eff}) were performed by using machine learning methods, respectively. The results indicate that incorporating GSD into the inversion process can help improve the accuracy of effective radius inversions.**

***Index Terms*—Aerosol remote sensing, geometric standard deviation (GSD), infrared hyperspectral, radiative transfer, size distribution.**

I. Introduction

As A key regulator of atmospheric environments, atmospheric aerosols profoundly influence the Earth's radiation balance [1], [2], [3]. On the one hand, aerosols can alter the global energy budget through their direct radiative effects (e.g., scattering and absorption, et al.) [4], [5], [6]; on the other hand, they play an important role in cloud microphysical processes as they serve as the condensation nuclei for cloud formation [7], [8]. Due to its importance, a comprehensive understanding of aerosol optical properties holds critical importance in fields such as climate science and cloud microphysical process [9], [10], [11].

Optical remote sensing is one of the important means to obtain aerosol properties [12]. Since the radius of most aerosol particles are comparable to the wavelengths of the visible/near-infrared (VIS/NIR) radiation, the mechanism of aerosol remote sensing is primarily established based on the scattering effects of aerosols [13], [14], [15], [16], [17]. Common remote sensing approaches include active systems such as lidar and passive instruments such as sun photometers and satellite sensors (e.g., MODIS, POLDER, OMI, TROPOMI) [18], [19], [20], [21], [22]. Among these methods, passive optical remote sensing techniques are generally constrained by the requirement of solar radiation; therefore, the retrieval of the aerosol should be implemented in the daytime [23], [24]. Hyperspectral infrared remote sensing technology, due to its unique spectral resolution capability, has been widely applied to the retrieval of temperature/humidity profiles and gas concentration [25], [26], [27], [28]. Because the infrared observation can be performed continuously during day and night, it can act as a critical complement to passive remote sensing in VIS/NIR band. In infrared radiative transfer simulations, the scattering effect by aerosol particles is generally neglected due to their small size parameters in thermal infrared band. However, recent research demonstrated that aerosols exhibit strong emission effects in thermal infrared bands, resulting in notable responses within hyperspectral signals [29]. Sokolik [30] found the spectral radiative response of mineral dust in infrared bands, demonstrating the importance of hyperspectral resolution in identification of aerosol components. Pierangelo et al. [31] further achieved the joint inversion of dust aerosol optical depth (AOD) and vertical height and revealed the potential of infrared hyperspectral remote sensing in resolving the 3-D structures of aerosol. Zhang et al. [32] constructed a spectral masking algorithm for dust storm events using thermal infrared channel, significantly improving the identification accuracy of dust spatial distribution. Gangale et al. [33] successfully realized the retrieval of the volcanic ash aerosol from hyperspectral infrared observations. Turner [34] achieved retrieval of AOD and effective radius (R_{eff}) from spectral infrared radiance observations in the 8–13 μm range, with the algorithm capable of differentiating three distinct mineral types. Ji et al. [35] employed the optimal estimation method to retrieve the optical depth and effective radius for different aerosol types based on infrared spectral measurements. Liu et al. [23] conducted an attempt to retrieve AOD using the

Received 30 August 2025; revised 17 December 2025; accepted 21 January 2026. Date of publication 26 January 2026; date of current version 17 February 2026. This work was supported in part by the Science Foundation of Hunan Province under Grant 2024JJ2058; and in part by the National Science Foundation of China under Grant 42175154, Grant 42205122, and Grant 462105367. *(Corresponding author: Shuai Hu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3657817

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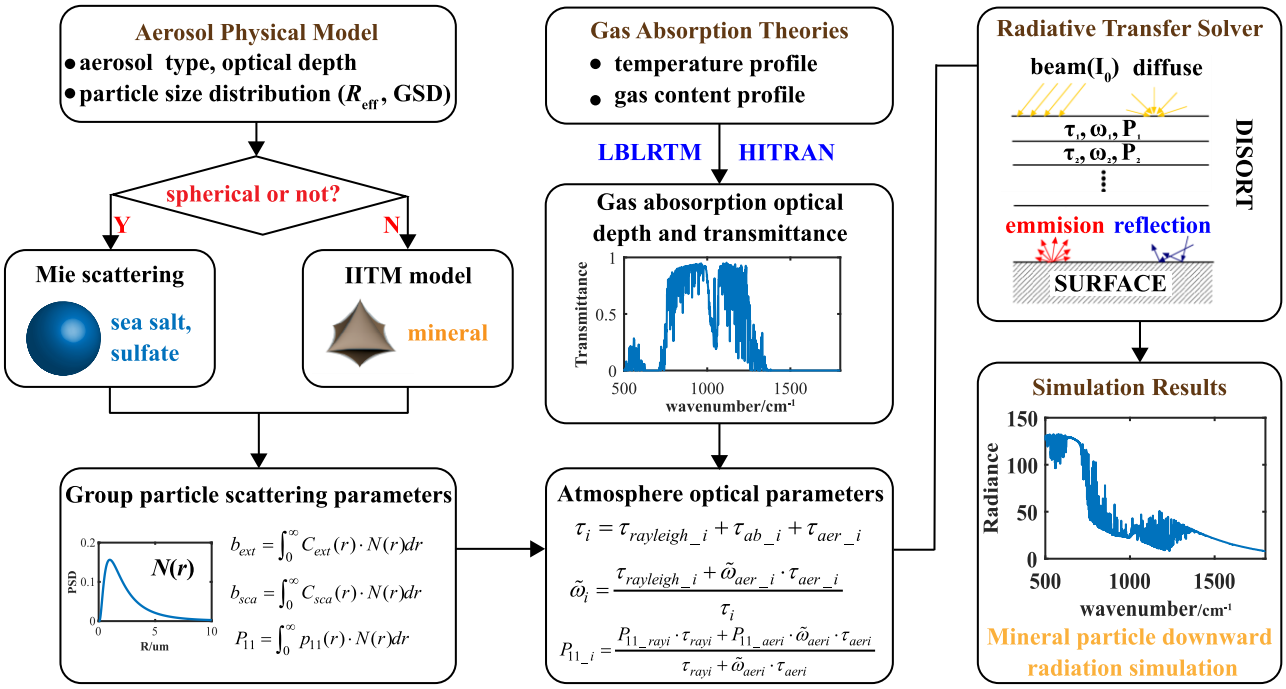


Fig. 1. Calculation frame of the hyperspectral infrared radiative transfer coupling the scattering and absorption process of aerosol particles.

atmospheric window channels, further validating the possibility and potential of the infrared hyperspectral remote sensing of aerosols in nighttime.

To explain their remote sensing mechanisms, an accurate analysis of the influence of the aerosol parameters on the infrared hyperspectral radiation is required. Among the microphysical parameters of aerosol, size distribution is a critical parameter characterizing aerosol optical properties, which determine extinction, scattering, and absorption properties of aerosol particles. In current studies, many mathematical models are applied to describe size distributions of aerosol, including gamma distribution, lognormal distribution, and so on [36]. Among these models, the lognormal distribution function has been widely adopted due to its mathematical simplicity and unambiguous physical interpretation [35]. This function is defined by two variables: R_{eff} and geometric standard deviation (GSD). However, in order to reduce the complexity in the retrieval of aerosol, previous studies generally assumed that the aerosol size distribution is solely dependent on R_{eff}, while GSD is fixed as a constant parameter. However, from recent studies, GSD generally changes with the environment and time. Palacios-Peña et al. [37] demonstrated that GSD could variate 1.11–2.80 from the measured data, and the spatial variability was closely linked to the sources and aging processes of aerosol. Unfer et al. [38] further investigated aerosol size distribution in the Amazon Basin, pointing out that GSD will fluctuate within the range of 1.39–1.50. Due to the evidence of substantial variability of GSD, some questions should be answered: how much does the fluctuation of GSD affect the infrared hyperspectral radiance? Is it as important as the R_{eff}? Is it necessary to take it into account in the modeling of the infrared hyperspectral transfer and the remote sensing of atmospheric aerosol?

To address this problem, the hyperspectral infrared radiative transfer modeling system coupling the scattering and absorption process is established, the influence of GSD on upwelling and downwelling hyperspectral radiance are quantified for different aerosol types and AODs. Furthermore, to further validate the necessity of considering GSD in aerosol remote sensing, both triple-parameter (AOD, R_{eff}, and GSD) and dual-parameter inversions (AOD and R_{eff}) were performed by using machine learning methods, respectively.

This article is organized as follows: in Section II, the hyperspectral infrared radiative transfer model system coupling the aerosol scattering and absorption process is established; in Section III, the influence of GSD and R_{eff} on the infrared radiance is systematically investigated and compared with each other. In Section IV, both triple-parameter (AOD, R_{eff} and GSD) and dual-parameter inversions (AOD and R_{eff}) were performed to further validate the necessity of considering GSD in aerosol remote sensing. Section V is a brief summary of this article.

II. Establishment of the Hyperspectral Infrared Radiative Transfer Model System Coupling the Aerosol Scattering and Absorption Process

To investigate the influence of GSD on the hyperspectral infrared radiances, we established a hyperspectral infrared radiative transfer modeling system by synergizing LBLRTM [39] and DISORT [40], as shown in Fig. 1. In this model system, the aerosol scattering and absorption processes were considered, and R_{eff} and GSD are set as the input variables of the model. The LBLRTM-DISORT chain achieves spectral radiance simulations with 0.01 cm^{-1} resolution, in which HITRAN2020 is applied to resolve molecular absorption effects, enabling precise calculation of optical depth at each atmospheric layer through

$$\tau = \sum_{j=1}^{N} \tau_j = \int_u \sum_{j=1}^{N} k_{v,j}(u) du \quad (1)$$

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TABLE I
AEROSOL SCATTERING PARAMETER DATABASE

Type	Insoluble, Water-soluble, Soot, Sea Salt, Mineral, Sulfate
Complex refractive index	Hess et al. (1998)
Shape	Spherical: Sea Salt, Sulfate, Water-soluble, Non-spherical: Insoluble, Mineral
wave bandwidth [cm^{-1}]	500-1800(dv=0.5)
particle radius [um]	0.0001-0.02(dv=0.001), 0.12-50(dv=0.1)

in which, τ is the optical depth, $j(= 1, 2, \ldots, N)$ represents the absorption coefficient index of the jth spectral line, v is the wavenumber, and $k_{v,j}$ represents the absorption coefficient of the jth spectral line.

The parameterization of the aerosol scattering process is an important part of the model. In this study, the single-particle scattering database was established by using Mie scattering code and T-matrix methods. The optical parameters in this database include extinction cross section ($C_{\text{ext,}}$) scattering cross section (C_{sca}), and scattering function p_{11}. The scattering property dataset encompasses 27 aerosol species, which are classified into six major types: insoluble aerosol (INSO), mineral dust (MICM), sea salt (SSAM), sulfate (SUSO), and water-soluble aerosol (WASO). To account for hygroscopic growth, the SSAM, SUSO, and WASO types each include subcategories at eight relative humidity levels (0%, 50%, 70%, 80%, 90%, 95%, 98%, and 99%), while INSO, MICM, and soot are considered nonhygroscopic. The particle diameter ranges from 0.0001 to 50 μm, the spectral resolution is 0.5 cm^{-1} between 500 and 1800 cm^{-1}. The main parameters of the aerosol are systematically tabulated in Table I.

Based on the single scattering database, ensemble-averaged scattering properties can be obtained through the integration over aerosol size distribution. In this model, the size distribution of aerosol is taken as the lognormal distribution, defined as follows:

$$n(r) = \frac{dN(r)}{dr} = \frac{N}{\sqrt{2\pi} r \log\sigma} \exp\left[-\frac{1}{2}\left(\frac{\log r - \log R_{\text{eff}}}{\log\sigma}\right)^2\right] \tag{2}$$

where $N(r)$ denotes the number concentration at radius r, N represents the total number concentration, r is the particle radius, R_{eff} is the effective radius, and σ is the GSD, where R_{eff} and GSD can be altered as the input parameters. By integrating single scattering parameters, the ensemble-averaged properties of aerosol particles in each atmospheric layer can be calculated as follows:

$$b_{\text{ext}} = \int_0^\infty C_{\text{ext}}(r) \cdot n(h, r) dr \tag{3}$$

$$b_{\text{sca}} = \int_0^\infty C_{\text{sca}}(r) \cdot n(h, r) dr \tag{4}$$

$$P_{11} = \int_0^\infty p_{11}(r) \cdot n(h, r) dr \tag{5}$$

in which, b_{ext}, b_{sca}, and P_{11} denote the extinction coefficient, scattering coefficient, and scattering function, respectively, and $C_{\text{ext}}(r)$, $C_{\text{sca}}(r)$, $p_{11}(r)$ denote the extinction, scattering cross section, and scattering function of single aerosol particles at different radius, respectively. Before the numerical simulation, P_{11} should be normalized by b_{sca} to obtain the scattering phase function.

By integrating the aerosol scattering parameters with the optical parameters of atmospheric molecules, layer-specific atmospheric parameters can be derived, including optical thickness (τ_i), single-scattering albedo (SSA) (ω_i), and scattering phase function (P_{11_i}). Finally, taking these layer-resolved parameters into the DISORT solver, infrared spectral radiance characteristics under varying aerosol size distributions can be calculated. Within the radiative transfer model, the vertical distribution of aerosol concentration $N(h, r)$ is assumed to follow an exponential law with altitude, written as follows:

$$N(h, r) = N(r) \cdot \exp(-h) \Big/ \int_0^{h_1} \exp(-h) dh \tag{6}$$

where $N(h, r)$ is the number density of aerosol particles with a radius of r at different heights, h the height above the Earth surface.

III. DISCUSSIONS OF THE INFLUENCE OF GSD AND REFF ON THE INFRARED RADIANCE

To quantify the impacts of aerosol size distribution on infrared radiance, a set of size parameter perturbation experiments are implemented based on the lognormal distribution function. By varying effective radius and GSD, we simulated upwelling and downwelling infrared hyperspectral radiances for five representative aerosol types, that is, dust, black carbon, sulfate, SSAM, and insoluble particles. Sensitivity of the radiance to the size parameters is described by the normalized relative variation (RV), written as follows:

$$\text{RV} = \frac{I_P - I_{\text{OPAC}}}{I_{\text{OPAC}}} \times 100\% \tag{7}$$

in which I_{OPAC} corresponds to the reference radiances simulated with standard OPAC size distributions (e.g., dust: GSD = 2.15, R_{eff} = 1.9 μm) [41], I_P represents the simulated upwelling/downwelling infrared radiances under varied size distributions.

Fig. 2 presents sensitivity of downwelling infrared radiance to GSD and R_{eff} for different aerosol types. In this simulation, the background atmospheric profiles were taken as U.S. standard atmosphere, with AOD set as 0.06 at 900 cm^{-1}. The size R_{eff} and GSD are perturbed by $\pm$10% and $\pm$20%, respectively, based on the standard OPAC size distributions. To facilitate the analysis of the physical mechanism of aerosols in infrared radiative transfer, SSA at 900 cm^{-1} under different particle size distributions (PSDs) were also calculated additionally, the results are shown in Tables II and III.

As can be seen, mineral aerosols exhibit notable sensitivity to variation of GSD, when the GSD disturbance is $\pm$20%, the RV of the radiance is as high as 8%, which is higher than the impact caused by the variation of R_{eff} (3%). The reason is that the infrared radiation is caused by the emission of mineral aerosols, which is closely dependent on the absorption

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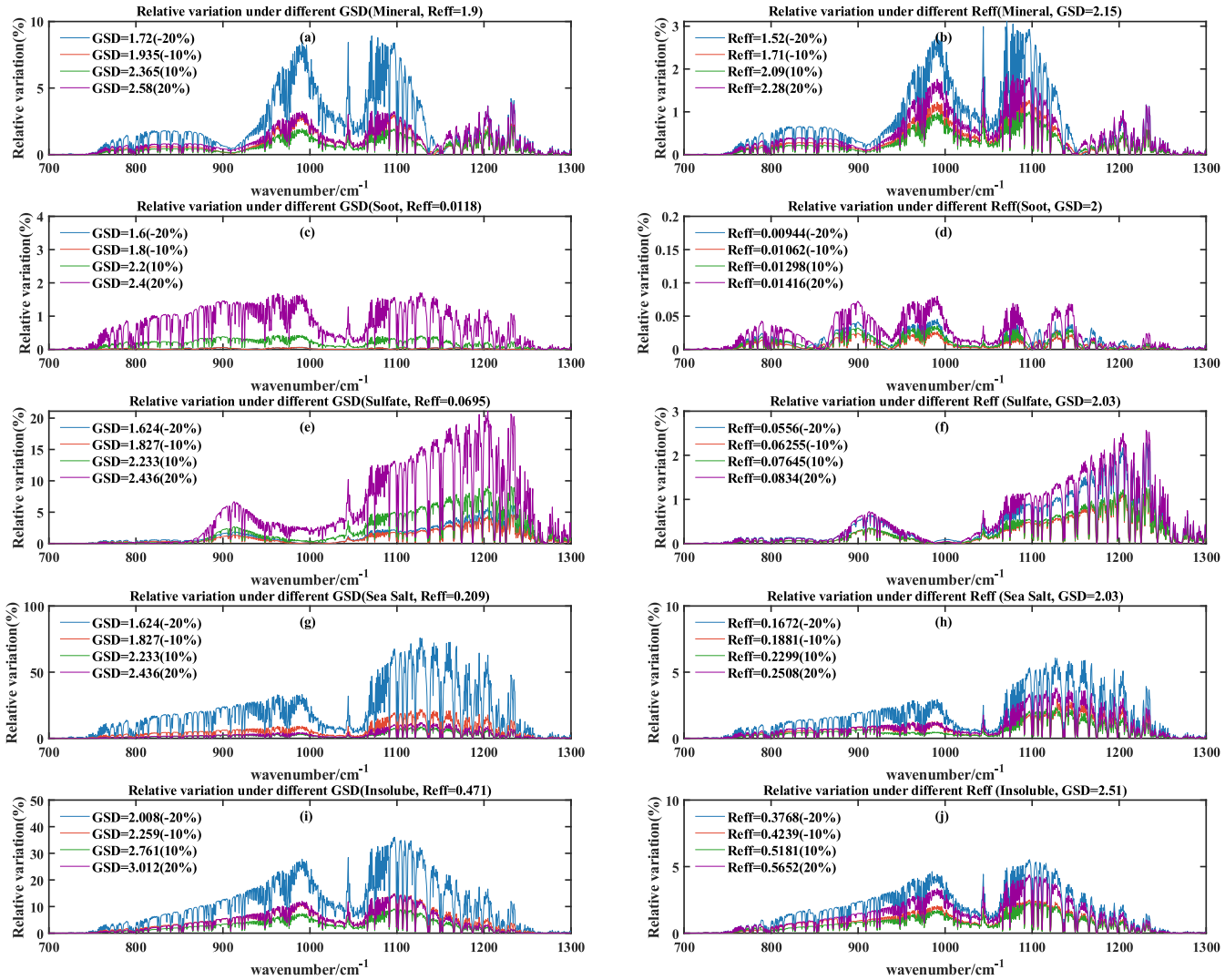


Fig. 2. (a) Relative variations of the downward radiation values of mineral under different GSD. (b) Relative variations of the downward radiation values of mineral under different Reff. (c) Relative variations of the downward radiation values of soot under different GSD. (d) Relative variations of the downward radiation values of soot under different Reff. (e) Relative variations of the downward radiation values of sulfate under different GSD. (f) Relative variations of the downward radiation values of sulfate under different Reff. (g) Relative variations of the downward radiation values of sea salt under different GSD. (h) Relative variations of the downward radiation values of sea salt under different Reff. (i) Relative variations of the downward radiation values of insolube aerosol under different GSD. (j) Relative variations of the downward radiation values of insolube aerosol under different Reff.

TABLE II

SINGLE SCATTERING ALBEDO OF AEROSOL WITH DIFFERENT GSD

Type	-20%	-10%	Standard PSD	+10%	+20%
Mineral	0.70	0.66	0.63	0.60	0.58
Soot	8.61E-6	9.17E-5	1.20E-3	8.00E-3	3.33E-2
Sulfate	2.80E-3	1.33E-2	4.97E-2	0.12	0.22
Sea Salt	0.63	0.85	0.93	0.95	0.96
Insolube	0.48	0.5	0.61	0.63	0.63

ability of the particles. According to the Rayleigh scattering theory, the absorption efficiency is proportional to the size parameter $x = 2\pi r/\lambda$, that is, Q_{ab}–x, namely, Q_{ab} is mainly determined by large particles. With the increase of GSD, the proportions of large particles will increase, thereby enhancing the absorption ability will be enhanced as well. To further provide a quantitative explanation for this phenomenon, we further analyze the variation of the single scattering albedo. As can be seen in Tables II and III, a 20% reduction in GSD from the reference value (GSD = 2.15) results in a decrease in mineral SSA from 0.70 to 0.63 (reduced by 11.1%), whereas equivalent perturbations of R_{eff} only reduce single scattering albedo from 0.65 to 0.63 (reduced by 3.2%). For soot aerosols, though the infrared radiance shows weaker sensitivity to size distribution parameters, the influence of GSD is stronger than effective radius on the whole, where the variation of the SSA at 900 cm^{-1} caused by a 20% variation of GSD and effective radius are about 99.3% and 62.5%, respectively. The phenomenon can be explained by the high absorptive ability and small size of soot, no matter how the size parameter changes,

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TABLE III
SINGLE SCATTERING ALBEDO OF AEROSOL WITH DIFFERENT R_{eff}

Type	-20%	-10%	Standard PSD	+10%	+20%
Mineral	0.65	0.64	0.63	0.62	0.61
Soot	4.50E-4	7.44E-4	1.20E-3	1.70E-3	2.30E-3
Sulfate	3.14E-2	4.02E-2	4.97E-2	5.98E-2	7.03E-2
Sea Salt	0.91	0.92	0.93	0.94	0.94
Insolube	0.60	0.61	0.61	0.62	0.62

the single scattering albedo is less than 0.05. Similar to mineral aerosol, the infrared radiance of sulfate, SSAM, and insoluble aerosols also show high sensitivity to size distribution, and the influence of GSD is notably larger than effective radius. The reason is similar, namely, the higher sensitivity of SSA to GSD than the effective radius R_{eff}, as shown in Tables II and III. These findings indicate that GSD is an important factor influencing the hyperspectral infrared radiative transfer, and it should be equally considered as the R_{eff} to accurately quantify radiative effect of the aerosol particles.

The influence of GSD and effective radius on the upwelling infrared radiance is also investigated. In this test, the background atmospheric profiles were also taken as the U.S. standard atmosphere, and AOD is set as 0.06 at 900 cm^{-1}. The sensitivity of the infrared radiance to the size parameters is also described by the normalized RV, and the results are shown in Fig. 3.

As can be seen, the size distribution's effects on upwelling radiance are generally weaker than those on downwelling radiance (RV is reduced by 1–2 magnitude orders for all aerosol types). This phenomenon can be explained by the fact that aerosol particles are mainly concentrated below the boundary layer. The downwelling radiance at the surface follows a short atmospheric path with limited interference, making it highly sensitive to aerosol properties. By contrast, the upwelling radiance at the top of the atmosphere undergoes a much longer path through the atmosphere, where aerosol signals are attenuated, leading to comparatively lower sensitivity. Nevertheless, GSD remains the dominant sensitivity parameter. For mineral aerosol, a 20% reduction of GSD from the reference value (GSD = 2.15) resulted in a change in upwelling radiance RV of 0.15%, whereas equivalent perturbations of R_{eff} only cause a variation of 0.06%. For sulfate, SSAM and insoluble aerosols, the sensitivity of upwelling radiance to PSD is also lower than that of downwelling radiance, but the sensitivity of upwelling radiance to GSD is higher than that of R_{eff} as well. For SSAM aerosols, when the GSD disturbance is $\pm$20%, the maximum RV of the radiance is as high as 0.5%, which is higher than the maximum RV caused by effective radius (0.06%). For soot aerosols, due to the small SSA, the sensitivity of upwelling radiance to size distribution parameters is also weak.

The impacts of GSD on infrared radiative are also investigated under different AODs. In this study, AOD is set to be 0.03, 0.06, and 0.09 at 900 cm^{-1}, and the background atmospheric profiles were also taken as U.S. standard atmosphere. The sensitivity of the radiance to the size parameters is also described by the normalized RV, the results are shown in Fig. 4.

As can be seen, the sensitivity of infrared radiation to the GSD will be amplified with the increase of AOD. For mineral aerosols [Fig. 4(a)–(c)], when the GSD disturbance is $\pm$20%, the maximum RV of the radiance is 5% under AOD = 0.03, but for AOD = 0.09, the RV is increased to 10% [Fig. 4(c)]. This phenomenon can be explained by the concentration of aerosol particles. With the increase of AOD, aerosol concentration will also become large, which leads to a greater impact of GSD. For sulfate, SSAM, and insoluble aerosols, they also exhibit greater sensitivity to the increase of GSD, where when AOD = 0.09, the sensitivity of downwelling radiance to GSD increased by 2–3 times compared with that for AOD = 0.03. Though soot aerosols show lower sensitivity to GSD compared with other aerosol types, the RV also increases with the increase of AOD. These findings indicate that the influence of GSD will become more and more remarkable with the increase of AOD, which further validates the importance of GSD in the parameterization of aerosol's optical properties.

IV. COMPARATIVE ANALYSIS OF DUAL-PARAMETER (AOD AND R_{Eff}) AND TRIPLE-PARAMETER INVERSION (AOD, R_{Eff}, AND GSD)

To further validate the importance of GSD and the necessity to consider GSD in hyperspectral infrared remote sensing of aerosol, both triple-parameter (AOD, R_{eff}, and GSD) and dual-parameter inversions (AOD and R_{eff}) were performed by the BP neural network method, respectively. The basic framework of the inversion model is shown in Fig. 5.

In the BP inversion model, the downwelling infrared hyperspectral radiance observed at the surface is taken as the input parameters, and the aerosol parameters are set as the output parameters, where, in triple-parameter, AOD, R_{eff}, and GSD are taken as the output, and in dual-parameter inversions, AOD and R_{eff} are retrieved. The BP neural network consists of two hidden layers, consisting of 120 and 80 neurons, respectively.

The training and validation datasets are established by using the radiative transfer model described in Section II. In this process, the aerosol type is set as dust particles, the downwelling infrared radiances were calculated under different aerosol parameters, where, AOD is ranging from 0.01 to 2 with an increment of 0.01, the variation region of R_{eff} is set as 0.01–2, and the increment is taken as 0.1; GSD is ranging from 1.15 to 3.15, and the step size is set as 0.1. From the numerical simulation, a total of 80 000 samples were generated, with 70% selected as the training set and the remaining 30% as the test set. During the inversion process, 150 feature channels (770–1250 cm^{-1}) were selected based on the maximum relevance and minimum redundancy algorithm [42].

To validate the effectiveness and robustness of inversion experiments, according to the 2025 edition of the AERI technical manual [43], a random noise of 1% was added into the hyperspectral infrared radiance across various bands. It is worth noting that this level is also within the range of radiometric noise observed in several spaceborne hyperspectral sounders (the HIRAS hyperspectral infrared atmospheric

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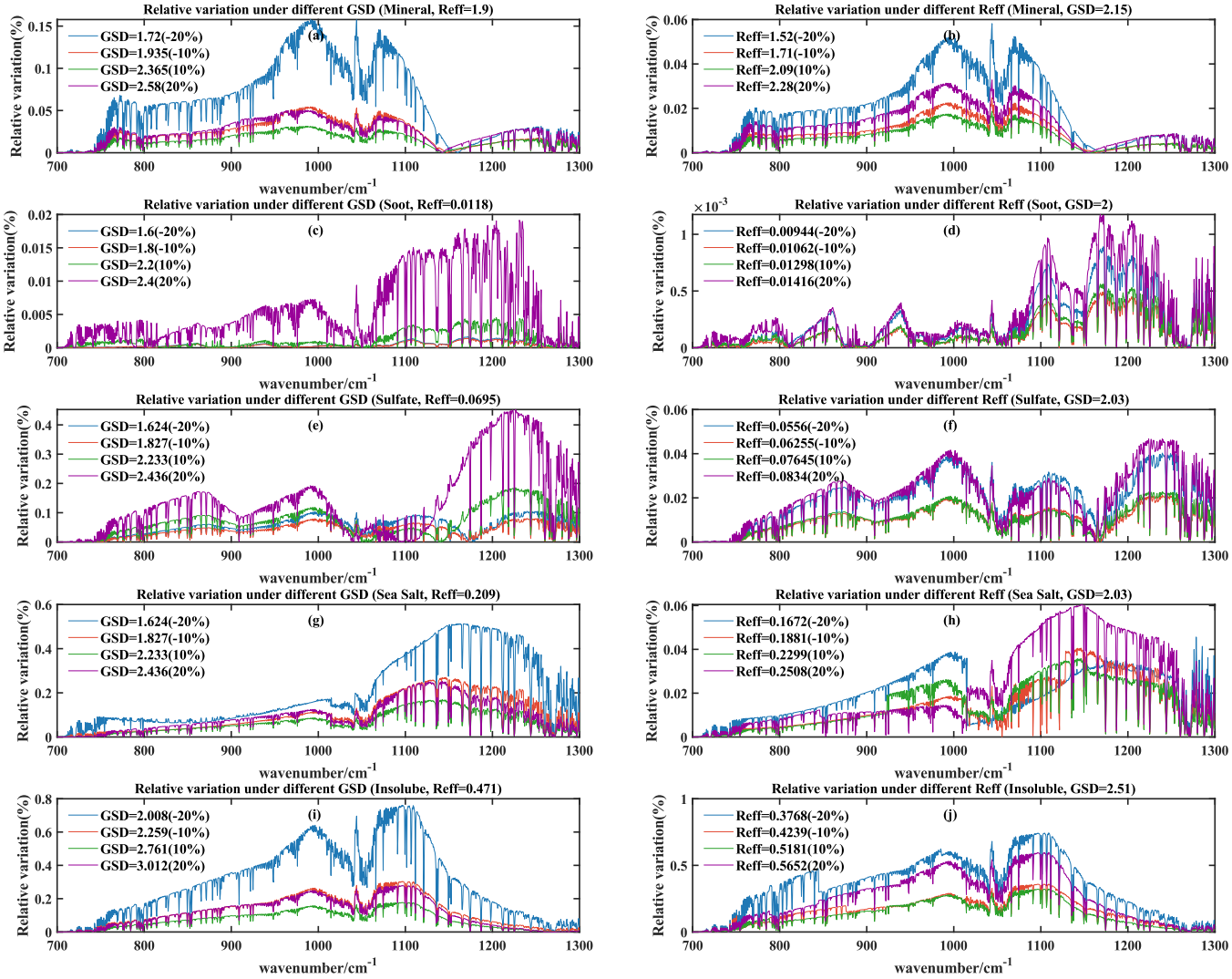


Fig. 3. (a) Relative variations of the upward radiation values of mineral under different GSD. (b) Relative variations of the upward radiation values of mineral under different Reff. (c) Relative variations of the upward radiation values of soot under different GSD. (d) Relative variations of the upward radiation values of soot under different Reff. (e) Relative variations of the upward radiation values of sulfate under different GSD. (f) Relative variations of the upward radiation values of sulfate under different Reff. (g) Relative variations of the upward radiation values of sea salt under different GSD. (h) Relative variations of the upward radiation values of sea salt under different Reff. (i) Relative variations of the upward radiation values of insolube aerosol under different GSD. (j) Relative variations of the upward radiation values of insolube aerosol under different Reff.

sensor aboard the FY series satellites features a radiometric calibration accuracy of 1 K, with a noise level below 1% relative to the measured data.), making it a representative value for a general simulation study. Dual-parameter inversion for AOD and R_{eff} was performed (Fig. 6), as well as triple-parameter inversion for AOD, R_{eff}, and GSD (Fig. 7).

The results indicate that the GSD can be successfully retrieved by the triple-parameter inversion scheme, and after incorporating the GSD as an additional retrieval parameter in the inversion process, the retrieval accuracy for R_{eff} is improved. The correlation coefficient for R_{eff} increased from 0.9237 to 0.9532 For AOD inversion, both approaches maintain high accuracy.

The root mean square error [RMSE, see (8)] of the inversion results from the two inversion schemes were also calculated separately, the results are shown in Table IV

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{M} \left(X_{\text{true}}(i) - X_{\text{retrieval}}(i)\right)^2}{M}}. \tag{8}$$

TABLE IV

RMSE FOR DUAL PARAMETER AND TRIPLE PARAMETER INVERSION SCHEMES AT THE NOISE LEVEL OF 1%

	AOD	R_{eff}	GSD
Dual-parameter	0.0374	0.2217	
Triple-parameter	0.0345	0.1750	0.1650

As can be seen, under the 1% noise level, the triple-parameter inversion scheme can not only achieve higher correlation coefficients but also exhibit comprehensively lower RMSE values compared with the dual-parameter inversion scheme. Specifically, the RMSE for R_{eff} was significantly reduced from 0.2217 to 0.1750. The error analysis of the retrieval results further confirms the superiority of the triple-parameter method.

To further validate the robustness of the inversion algorithm, the retrieval experiments are performed under higher noise

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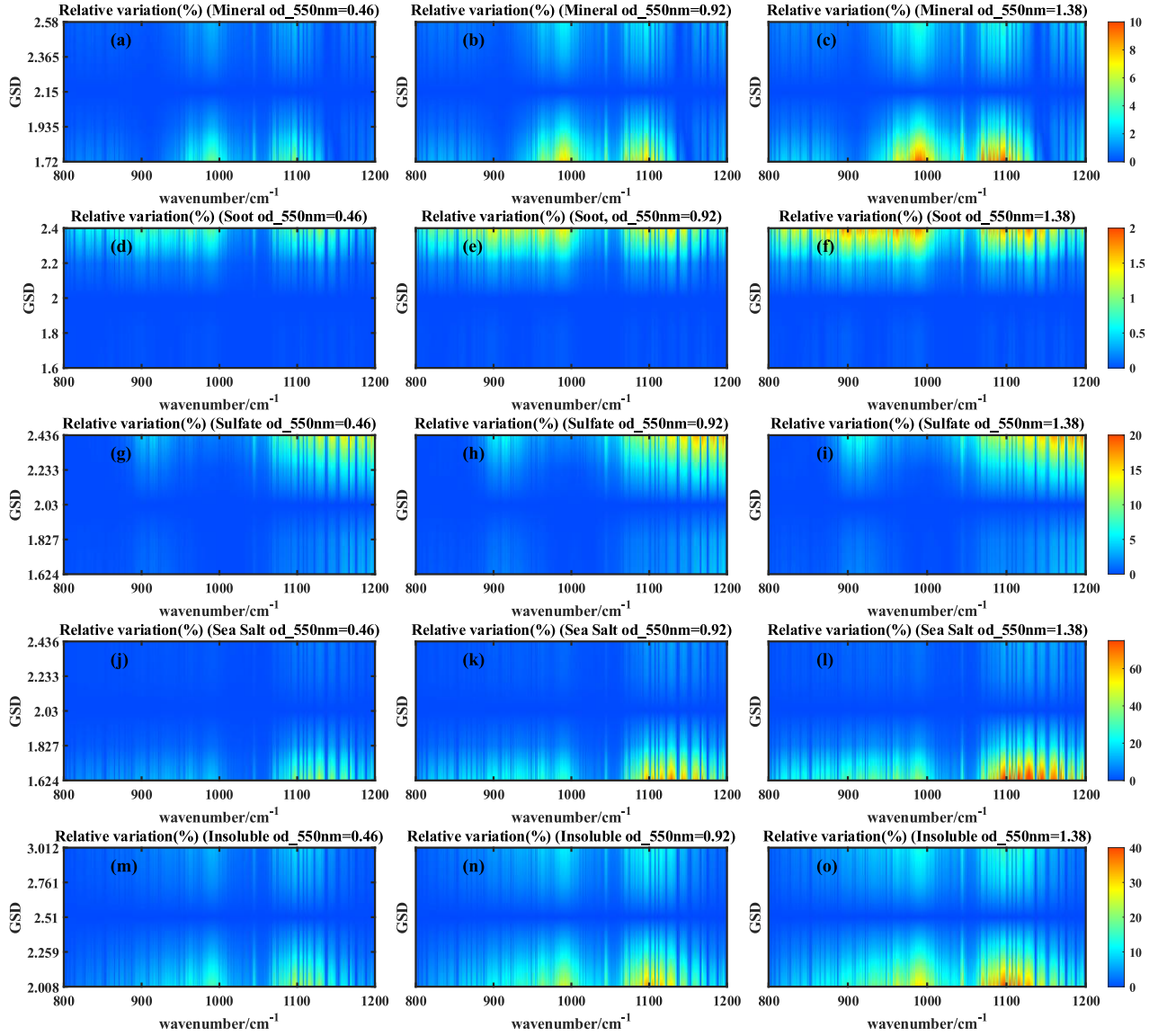


Fig. 4. (a) Relative variation of the downward radiation of mineral at an optical depth of 0.46. (b) Relative variation of the downward radiation of mineral at an optical depth of 0.92. (c) Relative variation of the downward radiation of mineral at an optical depth of 1.38. (d) Relative variation of the downward radiation of soot at an optical depth of 0.46. (e) Relative variation of the downward radiation of soot at an optical depth of 0.92. (f) Relative variation of the downward radiation of soot at an optical depth of 1.38. (g) Relative variation of the downward radiation of sulfate at an optical depth of 0.46. (h) Relative variation of the downward radiation of sulfate at an optical depth of 0.92. (i) Relative variation of the downward radiation of sulfate at an optical depth of 1.38. (j) Relative variation of the downward radiation of sea salt at an optical depth of 0.46. (k) Relative variation of the downward radiation of sea salt at an optical depth of 0.92. (l) Relative variation of the downward radiation of sea salt at an optical depth of 1.38. (m) Relative variation of the downward radiation of insolube aerosol at an optical depth of 0.46. (n) Relative variation of the downward radiation of insolube aerosol at an optical depth of 0.92. (o) Relative variation of the downward radiation of insolube aerosol at an optical depth of 1.38.

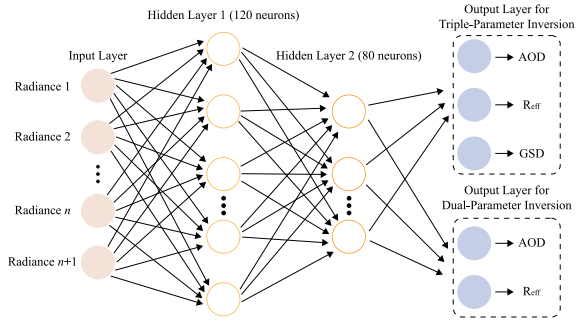


Fig. 5. Flowchart of neural network inversion scheme.

levels. In this study, a 2% measurement noise (according to the AERI Technical Manual 2025 edition) was introduced into the hyperspectral infrared radiance in 150 feature channels, and the performance of the dual-parameter and triple-parameter inversion methods are compared with each other, the results are shown in Figs. 8 and 9, respectively.

The results indicate that though the noise is increasing, the GSD can still be retrieved with high accuracy. Although the accuracy of both the two retrieval schemes decreased with the increase of the noise level, the triple-parameter inversion framework demonstrated its significant superiority and stability. As can been seen from the figures, under the influence of 2% noise, the correlation coefficients for R_{eff} retrieved by the triple-parameter method was 0.9217. Its accuracy remained superior to the results of the dual-parameter inversion (R_{eff}: 0.8681). By simultaneously retrieving the GSD with

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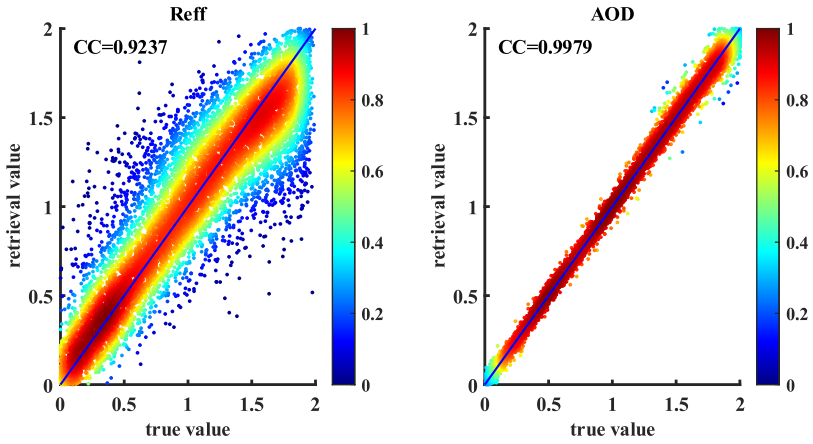


Fig. 6. Dual-parameter inversion results. The horizontal axis represents the true value, the vertical axis represents the retrieval value, and the noise setting is increased by 1%.

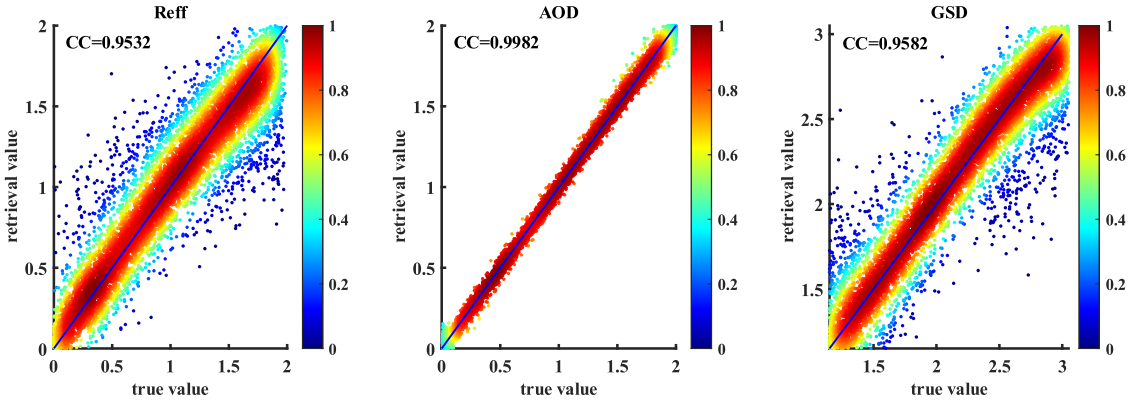


Fig. 7. Triple-parameter inversion results. The horizontal axis represents the true value, the vertical axis represents the retrieval value, and the noise setting is increased by 1%.

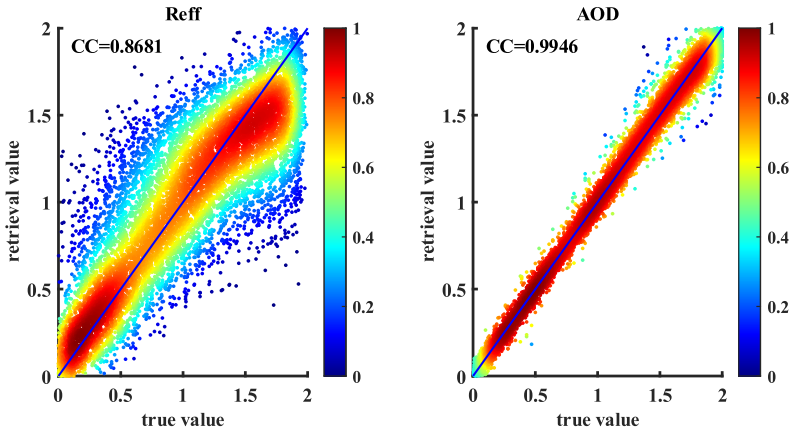


Fig. 8. Dual-parameter inversion results. The horizontal axis represents the true value, the vertical axis represents the retrieval value, and the noise setting is increased by 2%.

AOD and R_{eff}, it more fully utilizes the spectral information, thereby effectively constraining the solution space. Even in high-noise environments, it can yield more stable and accurate retrieval results for R_{eff}.

Similarly, the RMSEs of the dual-parameter inversion scheme and triple-parameter inversion scheme were also investigated under the noise level of 2%, and the results are shown in Table V.

From the table, it can be found that even in the higher noise level, the RMSEs of all parameters for the triple-parameter retrievals are lower than those for the dual-parameter inversions. Where, for AOD, the RMSE was reduced from 0.0599 to 0.0388 (a decrease of 35.2%), and for the inversion of R_{eff}, the RMSE also decreased from 0.2874 to 0.2246 (a decrease of 21.8%). The phenomenon further validates that the joint retrieval of GSD with AOD and R_{eff} can effectively improve the retrieval accuracy of each parameter overall. Besides, compared with Table IV, it can also be seen that with the increase of the noise level, the robustness of the triple-parameter inversion scheme is better than that of the

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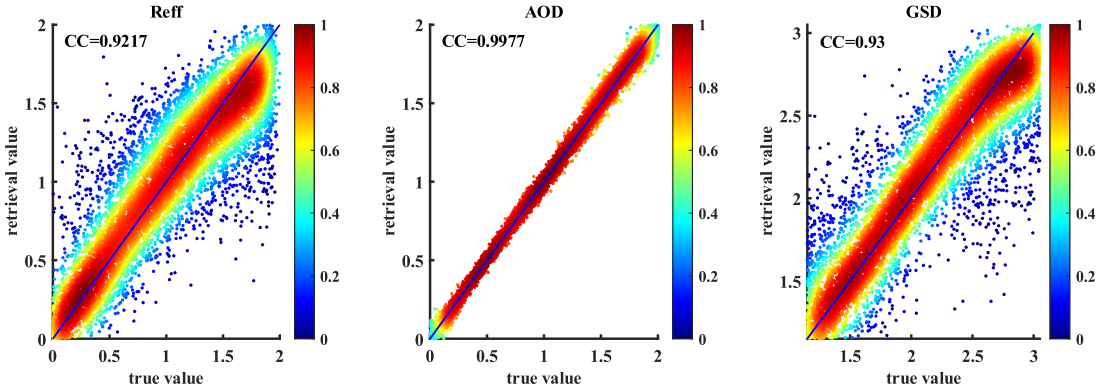


Fig. 9. Triple-parameter inversion results. The horizontal axis represents the true value, the vertical axis represents the retrieval value, and the noise setting is increased by 2%.

TABLE V
RMSE FOR DUAL PARAMETER AND TRIPLE PARAMETER INVERSION SCHEMES AT THE NOISE LEVEL OF 2%

	AOD	R_{eff}	GSD
Dual-parameter	0.0559	0.2874	
Triple-parameter	0.0388	0.2246	0.2123

dual parameter inversions. For the dual parameter inversion scheme, when the noise increases from 1% to 2%, the RMSE of AOD inversion increases by 0.0225. However, for the triple parameter inversion scheme, the RMSE of AOD inversion only increases by 0.0043.

V. CONCLUSION

Accurately analyzing the impacts of aerosol size parameter on infrared hyperspectral radiance is fundamental to elucidating their remote sensing mechanisms. However, previous studies generally assumed that the aerosol size distribution is solely dependent on effective radius, while GSD is fixed as a constant. This assumption neglects the influence of the width of size distribution on infrared radiative transfer and the remote sensing of aerosol. To address this problem, we quantify the influence of GSD on upwelling and downwelling hyperspectral radiance for different aerosol types and AODs. To further validate the importance of considering GSD in aerosol remote sensing, both triple-parameter (AOD, R_{eff}, and GSD) and dual-parameter inversions (AOD and R_{eff}) were performed. The main conclusions are drawn as follows.

1) For all five aerosol types, the discrepancies of the infrared radiance attributable to variations of GSD exceed those caused by R_{eff}. The reason is that in Rayleigh scattering region, the absorption efficiency is proportional to the size parameter ($x = 2\pi r/\lambda$); therefore, SSA demonstrates stronger dependence on GSD than R_{eff}, making it one of the dominant factors affecting infrared radiative transfer. Taking mineral aerosols as example, a 20% reduction in GSD results in an 11.1% decrease of SSA, whereas equivalent perturbations of R_{eff} only reduce SSA from 0.65 to 0.63 (3.2%). This confirms that a fixed GSD in size distribution will introduce systemic biases in simulation of the infrared radiance. Furthermore, when processing real measurements, simplistically assuming a fixed GSD value can become a significant source of retrieval error. Therefore, in hyperspectral infrared retrievals, it is necessary to treat GSD as a key variable or rigorously constrain its uncertainty to improve the accuracy of extracting aerosol microphysical properties from measured data.
2) The sensitivity of infrared radiation to the width of size distribution will be amplified with the increase of AOD. When AOD varied from 0.03 to 0.09, the GSD disturbance is ±20%, and the RV of the radiance was amplified from 5% to 10%. This phenomenon is caused by the increase of aerosol concentration when AOD rises, which leads to a greater impact caused by GSD. These analyses indicate that neglecting the influence of GSD will introduce unignored biases in hyperspectral radiance calculations, especially when the AOD is large.
3) The addition of GSD as a third inversion parameter can effectively improve the inversion accuracy of the R_{eff}, and it has been validated under different noise levels. At a noise level of 1%, the correlation coefficient of the R_{eff} varies from 0.9237 to 0.9532 when GSD is taken into the retrieval process. Even under 2% noise interference, the advantages of triple-parameter inversion are maintained. These data indicate that while both dual- and triple-parameter methods achieve high accuracy in retrieving AOD, incorporating GSD notably enhances the accuracy of effective radius retrievals.

The findings of this study may provide insights for improving aerosol inversion algorithms for current hyperspectral infrared sensors, such as those aboard the FY-series satellites. Specifically, the influence of GSD across spectral bands suggests that future algorithm development may need to simultaneously account for the effects of effective radius and GSD. Given that GSD uncertainty represents a potential source of error, a practical strategy is to treat it as a constraint variable in the inversion process rather than assuming a fixed value.

This article mainly emphasizes the importance and potential of considering GSD in infrared hyperspectral radiative transfer and aerosol remote sensing. However, due to this reason, in Section IV, we only take ground-based remote sensing of dust aerosols as an example for validation. In the future, we will conduct more systematic research on aerosol type identification, aerosol parameter remote sensing (including space-based and ground-based observation) by using the hyperspectral infrared radiance.

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